

A Artifact Appendix

A.1 Abstract

This artifact reproduces every quantitative result in the paper. The SDF-Edge framework models neural-network inference as a synchronous dataflow (SDF) graph and applies the CASAP (Capacity-constrained As-Soon-As-Possible) method to derive provably bounded buffers and static schedules. It regenerates Tables 2–8, the five machine-checkable proof certificates, the SDF3 comparison (with an exported SDF3-loadable XML graph for independent cross-verification), and the bare-metal Cortex-M7 validation. The core framework is pure Python 3 using only the standard library (no third-party dependencies) and completes the full software suite in under 30 seconds on a commodity machine. The optional on-device validation runs on the Renode emulator and needs no physical hardware.

A.2 Artifact check-list (meta-information)

- **Algorithm:** CASAP capacity-constrained SDF buffer scheduling; rate-consistency and repetition-vector computation; tiered formal verification with proof-certificate generation.
- **Program:** Python 3 framework (`src/`) and C firmware for STM32H743 (`firmware/`).
- **Compilation:** None for the Python framework. Optional firmware rebuild uses `arm-none-eabi-gcc`.
- **Binary:** A prebuilt STM32H743 firmware image (`sdf_edge_firmware.elf`) is included.
- **Model:** Five NN dataflow graphs — MCUNet, MobileNetV2, MobileNetV2-MultiRate, EfficientNet-B0, ViT-Tiny — as SDF graphs (`src/models.py`); all included.
- **Data set:** Seven SDF3 benchmark graphs, included in the archive (no external download).
- **Run-time environment:** Any OS (tested Windows 11, Ubuntu 22.04); CPython ≥ 3.8 (validated on 3.12.1), standard library only; optional Renode ≥ 1.14 .
- **Hardware:** Any x86-64 or ARM host. No GPU, no accelerator, no special hardware. The Cortex-M7 target is emulated in Renode (no board).
- **Execution:** Deterministic; no auto-tuning or randomized search. Part A runs in < 30 s.
- **Metrics:** Buffer size (bytes / KB) and reduction ratio; schedule length (firings); iteration time and FPS; verification time; on-device timing error and SRAM error.
- **Output:** Console tables; JSON proof certificates in `certificates/`; SDF3-loadable XML in `sdf3_comparison/`.
- **Experiments:** Executed via the provided Python scripts; each maps to a specific paper claim (see § Evaluation).
- **How much disk space required (approximately)?:** < 20 MB.
- **How much time is needed to prepare workflow (approximately)?:** < 5 minutes (install a Python interpreter; no packages to install).
- **How much time is needed to complete experiments (approximately)?:** < 1 minute for the full Part A suite.
- **Publicly available?:** Yes.
- **Code licenses (if publicly available)?:** Source-available for academic evaluation and research; the CASAP method is patent-pending (U.S. provisional) and all patent rights are reserved (see `LICENSE`).
- **Data licenses (if publicly available)?:** The included SDF3 benchmark graphs and NN dataflow models are distributed under the same archive license.
- **Workflow framework used?:** None; plain Python scripts.
- **Archived (provide DOI)?:** <https://doi.org/10.5281/zenodo.21521374>

A.3 Description

A.3.1 How to access

The artifact is permanently archived on Zenodo (open access) at <https://doi.org/10.5281/zenodo.21521374>, corresponding to release v1.0.0 of `alnemari-m/sdf-edge`. Download and unpack the archive; no account or credentials are required.

A.3.2 Hardware dependencies

None. Any commodity x86-64 or ARM machine suffices; no GPU or special hardware. The Cortex-M7 measurements are produced on the Renode STM32H743 model, so no physical development board is required.

A.3.3 Software dependencies

Required (Part A): CPython ≥ 3.8 ; the framework imports only the Python standard library (`requirements.txt` lists no third-party packages). Optional (Part B): Renode ≥ 1.14 to run the firmware, and `arm-none-eabi-gcc` only if rebuilding the firmware from source.

A.3.4 Data sets

All input graphs (the five NN dataflow models and the seven SDF3 benchmark graphs) are contained in the archive. There are no proprietary or external data dependencies.

A.3.5 Models

The five neural-network dataflow models are defined programmatically in `src/models.py`; running the experiment driver emits a proof certificate for each under `certificates/`.

A.4 Installation

No installation is required beyond a Python 3 interpreter:

```
python --version          # expect 3.8+ (tested 3.12.1)
pip install -r requirements.txt # succeeds; installs nothing (stdlib only)
```

Smoke test (should print Ratio: 10.3x):

```
cd src && python buffer_proof.py
```

A.5 Experiment workflow

From the `src/` directory:

```
python run_experiments.py    # Tables 2-8 + regenerate all certificates
python buffer_proof.py       # headline CASAP vs ASAP buffer result
python sdf3_comparison.py    # comparison vs SDF3 + export SDF3 XML
python sdf3_benchmark_validation.py # validation on 7 SDF3 benchmark graphs
```

Optional on-device validation, from `firmware/`:

```
renode sdf_edge_sim.resc      # boots STM32H743 model, runs firmware
```

A.6 Evaluation and expected results

Each command reproduces a specific paper claim:

- **10.3× buffer reduction** (CASAP vs ASAP), 26-actor MobileNetV2-MultiRate: `buffer_proof.py` prints CASAP: 1.5 KB, ASAP: 16.0 KB, Ratio: 10.3x.

- **2.4× over SDF3’s best** capacity-constrained result, plus an SDF3-loadable XML export: `sdf3.comparison.py`.
- **SDF3 benchmark sweep:** CASAP improves on **5/7** graphs (1.5×–18.4×): `sdf3.benchmark_validation.py`.
- **Tables 2–8 and the five proof certificates:** `run_experiments.py`. Regenerated `certificates/*.json` are bit-stable versus the committed files.
- **On-device (Cortex-M7):** 5.3× reduction on the 8-actor model, < 7.4% timing error, < 2.4 KB SRAM error; reference captures in `firmware/uart_output.txt` and `firmware/renode_output.log`.

A.7 Experiment customization

New models are added by describing their dataflow graph in `src/models.py`; the framework then derives the repetition vector, $q_{\max}$, CASAP/ASAP buffer bounds, a static schedule, timing, and a proof certificate automatically. The exporter in `sdf3_comparison.py` emits standard SDF3 XML that loads directly into the independent `sdf3analysis-sdf` tool for third-party verification.

A.8 Notes

The zero-dependency, hardware-free design of Part A is deliberate: it guarantees compatibility with any evaluation host and makes the results reproducible in seconds. The code is provided for academic evaluation and research; the CASAP scheduling method is the subject of a pending U.S. provisional patent application and all patent rights are reserved (see `LICENSE`). Evaluation and reproduction of the artifact are expressly permitted.

A.9 Methodology

Submission, reviewing, and badging methodology:

- <https://www.acm.org/publications/policies/artifact-review-badging>
- <http://cTuning.org/ae/submission-20201122.html>
- <http://cTuning.org/ae/reviewing-20201122.html>